

Green Machines + Metallic materials

Year 9 Science · exam sprint · 2 hours

- 0:00 Lesson 1 · Green Machines
- 0:50 Lesson 2 · Metallic materials
- 1:40 Mini mock (both units)

Teaching line, not a question slide.
两小时考前冲刺，一对一，投屏远距离。学生基础好，节奏可以快。
时间表：Lesson 1 Green Machines 0:00–0:50；Lesson 2 Metallic materials 0:50–1:40；Mini mock 1:40–1:57；作业 1:57–2:00。
开课前发纸：Classwork (Part A–F) 和 Mini mock 先发 Classwork，Mini mock 到 1:40 再发；Homework 两份下课时发。全部下课收回。
评分语言统一用 A / M / E：A 写出事实，M 把原因连上，E 把整条链写完整并用数据或对比。
课件页脚的页码指她的两本 booklet：Green Machines p.X 和 Metals p.X。

LESSON 1 · Green Machines

Learning intention: explain how a plant makes food, moves water and makes new plants, using the exam words.

Teaching line, not a question slide.

时间 0:00–0:02。

告诉她这一小时的三段：Loop 1 reproduction，Loop 2 photosynthesis and respiration，Loop 3 transport。每段结尾 2–3 道 Practice，最后 10 分钟做 Classwork Part A–C。

她这本 booklet 基本都写了，所以今天不是重新教，而是把答案写到 M 和 E。

Say it:

photosynthesis = foh-toh-SIN-thuh-sis

respiration = res-pih-RAY-shun

Do Now

- 1 Which part of a flower makes pollen?
- 2 The pollen nucleus fuses with the ovule nucleus. Is this pollination or fertilisation?
- 3 Name the three conditions a seed needs to germinate.
- 4 Which gas does a leaf take in to make glucose?
- 5 Which tubes carry water up the stem?

Q1 anther

Q2 fertilisation

Q3 water, warmth, oxygen

Q4 carbon dioxide

Q5 xylem

Short solution:

Q1 anther 在 filament 顶端，两者合起来是 stamen

Q2 pollen 从 anther 到 stigma 是 pollination；两个 nucleus 融合才叫 fertilisation

Watch for:

Q2 答 pollination 说明两个词混了，这是她最容易丢的定义分

Q5 答 “veins” 不够，要写 xylem

Diagnostic:

Q2 错：Loop 1 的 “From pollen to seed” 多花 1 分钟

Q4 答 oxygen：把 photosynthesis 和 respiration 的方程式记反了，Loop 2 先讲方程式

Q5 答 phloem：xylem 和 phloem 混淆，Loop 3 的 Quick review 先做

Say it:

pollination = pol-ih-NAY-shun

fertilisation = fer-tih-ly-ZAY-shun

ovule = OV-yool

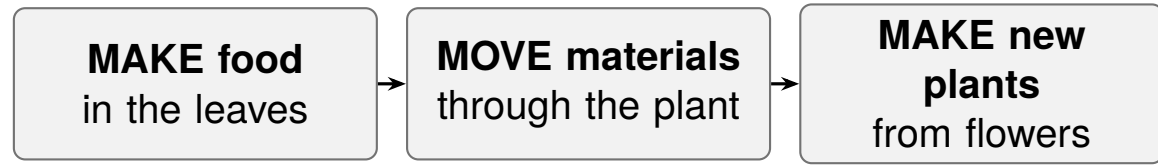
germinate = JER-mih-nayt

xylem = ZY-lum

phloem = FLOH-um

stamen = STAY-mun

The one idea



Every Green Machines question asks about one of these three jobs.
Say which job it is, then use that job's words.

Teaching line, not a question slide.

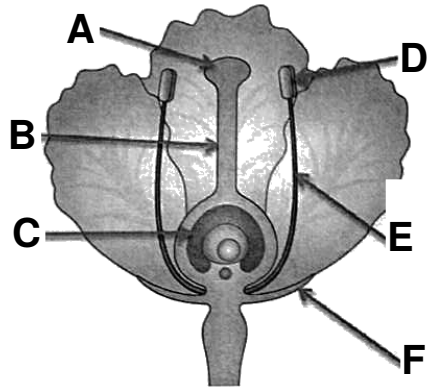
时间 0:02–0:03。

MAKE food = photosynthesis (叶子) ; MOVE materials = transport: water 走 xylem, glucose 走 phloem; MAKE new plants = reproduction: flower、seed、germination。

问她：“testing a leaf for starch 属于哪一个？” 答：MAKE food，因为 starch 是储存起来的 glucose。

考试时先判断是哪一件事，再用那一块的术语，答案就不会跑题。

Quick review: Parts of a flower



- 1 Name parts A to F.
- 2 Which two parts make up the stamen?
- 3 Which three parts make up the carpel?
- 4 Which part becomes the fruit?

1. A stigma, B style, C ovary, D anther, E filament, F sepal

2. anther and filament (D and E)

3. stigma, style and ovary (A, B and C)

4. the ovary (C)

Short solution:

4. fertilisation 之后 ovary wall 变成 fruit, 里面的 ovule 变成 seed

Watch for:

1. sepal 和 petal 不要混: sepal 在花的最外面最下面, 花还是花苞时保护它

3. carpel 也叫 pistil, 两个名字都可以

Say it:

sepal = SEE-pul

carpel = KAR-pul

Two ways to make new plants

	Asexual	Sexual
Parents	_____	two
Gametes?	_____	pollen + ovule
Offspring	_____	_____
Examples	runners, bulbs, cuttings	_____

A spider plant grows plantlets on long stems.
Which type of reproduction is this? How do you know?

Teaching line, not a question slide.
时间 0:05–0:07。
空格答案：Asexual 一栏 one parent / no gametes / identical (clones)；Sexual 一栏 offspring show variation / flowers and seeds。
问题答案：asexual reproduction，因为只有 one parent，没有 gametes，plantlets 和 parent plant identical (clones)。
她在 booklet p.4 把 “This specific process involves gametes” 也打了勾，这是 L2：asexual 一定没有 gametes。
Say it:
asexual = ay-SEK-shoo-ul
gametes = GAM-eets

Why variation matters: write it in this order

- 1 The environment changes, e.g. a new disease arrives.
- 2 Offspring from sexual reproduction are _____ from each other.
- 3 Some offspring have features that let them _____.
- 4 These survive and _____, so the species _____.

A: lines 1–2 **M:** links 2 to 3 **E:** the whole chain + the asexual contrast

Why could a whole field of identical plants die from one disease?

Teaching line, not a question slide.

时间 0:07–0:08。

空格: different (show variation) / survive the change (resist the disease) / reproduce / continues。

问题答案: the plants are clones with the same genetic material, 所以如果一株会得这种病, 所有植株都会得, 没有一株有抵抗力。

这是 Explain 长答题的标准链条。她 booklet p.7 写 “helps the species to live”, 只到 A。要拿 E 必须写出 “some survive → reproduce → species continues”, 并且和 asexual 对比。

From pollen to seed

Put the cards in order.

- A** The pollen tube grows down the style.
- B** Pollen lands on the stigma.
- C** The ovule becomes a seed; the ovary becomes a fruit.
- D** The pollen nucleus fuses with the ovule nucleus.
- E** The pollen tube enters the ovule through the micropyle.
- F** An insect carries pollen away from an anther.

Which cards are pollination? Which card is fertilisation?

Teaching line, not a question slide.

时间 0:08–0:10。

顺序：F, B, A, E, D, C。

Pollination = F 和 B (pollen 从 anther 到 stigma) 。 Fertilisation = D (两个 nucleus 融合，形成 zygote) 。

提醒 L1：pollen 落在 stigma 上还不是 fertilisation。

D 之后 zygote 长成 embryo，在 seed 里面。

Say it:

micropyle = MY-kroh-pile

zygote = ZY-goht

embryo = EM-bree-oh

Wind or insect pollination?

Part	Wind-pollinated	Insect-pollinated
Petals	small, dull, no scent	_____
Stigma	_____	sticky, inside the flower
Anthers	hang outside the flower	_____
Pollen	_____	larger, sticky or spiky

Full marks: the feature **and** how it helps.

Why do the anthers of grass hang outside the flower?

Teaching line, not a question slide.

时间 0:10–0:12。

空格：large, brightly coloured, scented, with nectar / large and feathery, hanging outside / firmly attached inside the flower / small, light, made in huge amounts。

问题答案：so the wind can blow the pollen away from the anthers。只写 “they hang outside” 是 A；写出 “so the wind can carry the pollen” 才是 M。

这是 L3：每个 feature 后面都要加 “so that ...”。

Spot the error (1)

A strawberry plant makes new plants from runners and from its flowers. Compare the two.

Sam's answer

Runners: the new plant gets gametes from the parent, so it is a bit different from it. Flowers: pollen is carried from the stigma to the anther. This is called fertilisation. The seeds show variation because two parents are involved.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. runners have no gametes: the new plant is identical to the parent, a clone (L2)
2. pollen is carried from the anther to the stigma, not the other way (L1)
3. that transfer is pollination, not fertilisation (L1)

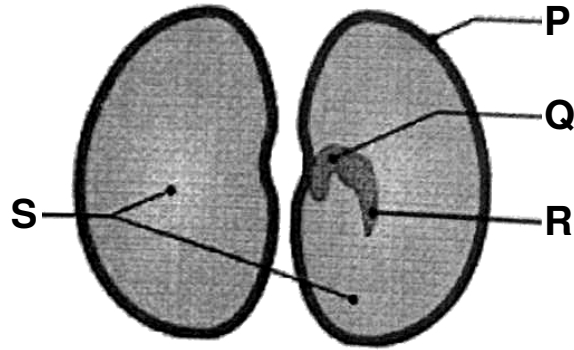
Short solution:

Runners are asexual: one parent, no gametes, so the new plants are identical to the parent. Flowers: pollen is carried from the anther to the stigma (pollination); the pollen nucleus fuses with the ovule nucleus (fertilisation). The seeds show variation because two parents are involved.

Watch for:

最后一句是对的。让她先判断每一句，而不是看到 Sam 就全部改掉。

Quick review: Seeds



- 1 Name parts P to S.
- 2 Name the three conditions for germination.
- 3 Why are seeds dispersed far from the parent plant?

1. P testa, Q plumule, R radicle, S cotyledon
2. water, warmth, oxygen
3. so the new plant does not compete with the parent for light, water, minerals and space

Short solution:

1. testa 是保护种子的硬外皮, water 让它变软; radicle 向下长成 root, 固定植物并吸水; plumule 向上长成 shoot; cotyledon 是 food store, 在叶子能 photosynthesise 之前提供能量
2. oxygen 用于 respiration, 释放生长需要的能量

Watch for:

1. booklet 印的是 “Radical”, 考试要拼 radicle
3. 只写 “to spread out” 是 A; 要写出 competition 和资源名称

Say it:

plumule = PLOO-myool

radicle = RAD-ih-kul

cotyledon = kot-ih-LEE-dun

dispersed = dis-PERST

Practice

Q1 A lily has large white petals, a strong scent and nectar. Is it wind- or insect-pollinated? Give two reasons. **[A]**

Q2 A grower takes cuttings from her best rose bush instead of planting its seeds. Explain why. **[M]**

Q3 Cress seeds are put in four tubes at 20 °C: A on dry cotton wool; B on wet cotton wool; C on wet cotton wool in a fridge; D under boiled water with a layer of oil on top. Which tubes germinate? Explain the result for D. **[E]**

Q1 insect-pollinated: large white petals and scent attract insects; nectar is food for insects

Q2 cuttings are asexual, so the new plants are identical to the best bush and keep its features; seeds come from sexual reproduction and show variation

Q3 only B germinates; D has no oxygen, so the seed cannot respire to release energy for growth

Short solution:

Q3 A 没有 water; C 没有 warmth (fridge 里温度太低) ; D: boiling 把水里溶解的 oxygen 赶走, oil 层不让空气重新溶进去

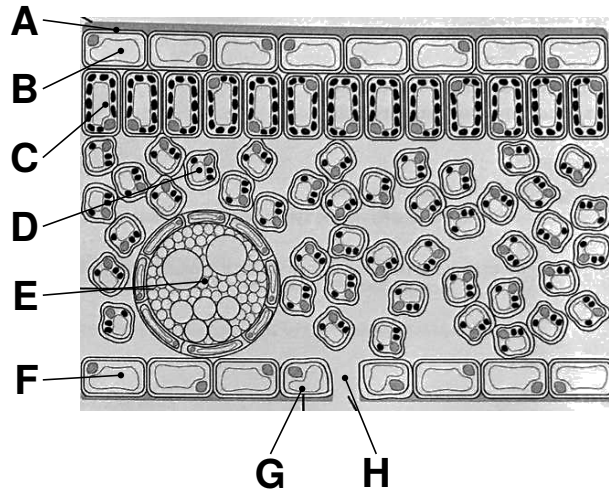
Watch for:

Q1 白色花瓣也算 “large, showy”, 不一定要彩色

Q3 C 的标准答案写 “too cold / no warmth”。D 一定要写 oxygen 的用途 (respiration 释放能量), 否则只有 M

Q3 这个 boiled water + oil 的设计和 Lesson 2 的 rusty nail 完全一样, 等一下会用到

Quick review: Leaf structure



- 1 Name A to H.
- 2 Which layer does most of the photosynthesis?

1. A waxy cuticle, B upper epidermis, C palisade layer, D spongy layer, E vein (xylem and phloem), F lower epidermis, G guard cell, H stoma

2. the palisade layer (C)

Short solution:

2. palisade cells 又高又窄、排得很紧，在叶子上部，含最多 chloroplasts

Watch for:

1. H 的复数是 stomata; 拼写常错

1. 她 booklet p.15 把 “transports water and minerals” 填成 spongy layer。应该是 vein 里面的 xylem (L7)

Say it:

cuticle = KYOO-tih-kul

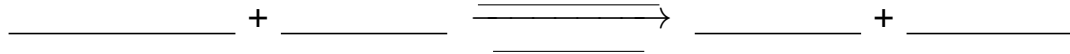
epidermis = ep-ih-DER-mis

palisade = pal-ih-SAYD

stomata = STOH-muh-tuh

chloroplasts = KLOR-uh-plasts

Photosynthesis: the word equation



Where does each starting substance enter the plant?

carbon dioxide: through the _____ in the leaf

water: root hairs → _____ → leaf

What does chlorophyll do in this reaction?

Teaching line, not a question slide.

时间 0:19–0:21。

方程式: carbon dioxide + water → glucose + oxygen; 箭头上写 light energy, 下面写 chlorophyll。

carbon dioxide 从 stomata 进入; water 从 root hairs 进入 (osmosis), 经 xylem 到叶子。

chlorophyll: absorbs light energy (真实的科学)。booklet p.20 写的是 chlorophyll “speeds up the reaction”, 按学校的说法两种都可以接受。

L4: light 和 chlorophyll 不是反应物, 写在箭头上下。

Say it:

chlorophyll = KLOR-uh-fil

osmosis = oz-MOH-sis

Respiration: the same substances, reversed

glucose + _____ $\longrightarrow$ _____ + _____ (+ energy)

	Photosynthesis	Respiration
Day	_____	_____
Night	_____	_____

Which process happens in every living cell, all the time?

Teaching line, not a question slide.

时间 0:21–0:23。

方程式：glucose + oxygen $\rightarrow$ carbon dioxide + water (+ energy)。在 mitochondria 里进行。

表格：Day 两个都 active；Night photosynthesis not active, respiration active。

问题答案：respiration。

L5：植物白天也 respire，只是白天 photosynthesis 更快，所以白天净放出 oxygen。

Say it:

mitochondria = my-toh-KON-dree-uh

Testing a leaf for starch

Give the reason for each step.

- 1 Put the leaf in boiling water for 1 minute.
- 2 Put it in hot ethanol, standing in a beaker of hot water.
- 3 Rinse it in water.
- 4 Add iodine solution. Blue-black means _____.

Why is the ethanol heated in hot water, not over a Bunsen flame?

Teaching line, not a question slide.

时间 0:23–0:25。

1. kills the leaf and stops its reactions (so the starch stays)
2. removes the chlorophyll, so the colour change can be seen
3. softens the leaf and washes off the ethanol
4. starch is present; 没有 starch 时保持 brown / yellow-brown

问题答案: ethanol is flammable, 不能靠近明火。

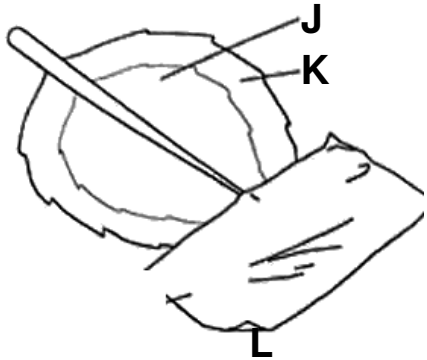
实验题常考 “why each step”, 她最容易只写步骤不写原因。

Say it:

ethanol = ETH-uh-nol

iodine = EYE-uh-dyne

Workbook: the foil leaf



A variegated leaf: J green, K white.
Foil L covered part of it for 2 days in the light. Then the leaf was tested with iodine.

Which areas turn blue-black? Explain each area.

(a) J (green, not covered): blue-black

(b) K (white): stays brown

(c) L (under the foil): stays brown, even where it is green

Short solution:

J 有 chlorophyll 也有 light, 进行 photosynthesis, glucose 变成 starch 储存

K 没有 chlorophyll, 不能 photosynthesise, 没有 starch

L 没有 light, 不能 photosynthesise, 没有 starch

Watch for:

这是 booklet p.22 第 2 题, 她没写。每个区域都要说“缺什么”: K 缺 chlorophyll, L 缺 light

Say it:

variegated = VAIR-ee-uh-gay-tid

Spot the error (2)

Write the word equation for photosynthesis. Then say how you would show that a leaf has made food.

Sam's answer

oxygen + water makes glucose + carbon dioxide (light above the arrow). Boil the leaf, put it in hot ethanol, rinse it, then add iodine. If it goes blue-black, the leaf has made glucose. Plants photosynthesise in the day and respire at night.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. carbon dioxide + water → glucose + oxygen: Sam swapped oxygen and carbon dioxide (L4)

2. blue-black shows starch, not glucose (L6)

3. plants respire all the time, day and night (L5)

Short solution:

carbon dioxide + water → glucose + oxygen, with light energy above the arrow and chlorophyll below. Blue-black with iodine shows the leaf contains starch. Plants respire all the time; they photosynthesise only in the light.

Watch for:

实验步骤那一句是对的。

Practice

Q1 Two potted plants are kept in the light for 2 days, each under a sealed bell jar. Jar 1 also holds a dish of soda lime, which absorbs carbon dioxide. Jar 2 holds a dish of water. A leaf from each plant is tested with iodine. Predict and explain the results. **[M]**

Q2 In a greenhouse full of plants, the carbon dioxide level rises at night and falls during the day. Explain why. **[E]**

Q1 plant 1 stays brown (no starch); plant 2 turns blue-black (starch)

Q2 at night only respiration happens, which gives out carbon dioxide; in the day photosynthesis takes in carbon dioxide faster than respiration gives it out

Short solution:

Q1 jar 1 没有 carbon dioxide, 不能 photosynthesise, 不能做 glucose, 所以没有 starch; jar 2 有 carbon dioxide, 是 control

Q2 要写清两个过程, 以及白天 “photosynthesis is faster than respiration”

Watch for:

Q1 她 booklet p.21 只写了 “there won't be photosynthesis”, 没写原因 (缺 carbon dioxide), 只能拿 A

Q2 写 “plants respire at night” 只答了一半; 只写 “plants breathe out carbon dioxide” 不给分

Quick review: Transport words

Match each word to its job.

- | | |
|----------------------|---|
| 1 xylem | A carries glucose from the leaves |
| 2 phloem | B lets gases and water vapour in and out of the leaf |
| 3 root hair | C absorbs water from the soil |
| 4 stomata | D carries water and minerals up the plant |
| 5 guard cells | E change shape to open and close the stomata |

1. D
2. A
3. C
4. B
5. E

Short solution:

记法: xylem 运 water 向上; phloem 运 food (glucose) 到需要或储存的地方

Watch for:

2. phloem 的拼写: ph 开头, -oem 结尾

5. booklet p.16 说 guard cells “swell to close” stomata。真实情况是 guard cells 吸水 swell 时 stomata 打开。考试如果问 swell 和 open 的关系, 按 booklet 写, 并提醒她两种说法都见过

How water travels through a plant

soil water

↓ by _____

root hair cell

↓

_____ in the root and up the stem

↓

leaf cells: water evaporates

↓

out through the _____ as water vapour

What is the loss of water vapour from the leaves called?

Teaching line, not a question slide.

时间 0:31–0:33。

空格： osmosis / xylem / stomata。

问题答案： transpiration。

Say it:

transpiration = tran-spih-RAY-shun

Why the water moves up: write it in this order

A celery stalk stands in blue dye for 2 days. The leaves turn blue.

- 1 Water _____ from the leaves through the stomata.
- 2 This _____ more water up the _____.
- 3 The dye travels with the water, so the _____ turn blue.
- 4 Cut the stalk: the blue dots show where the _____ are.

A: dye goes up **M:** names xylem **E:** transpiration causes the pull

Teaching line, not a question slide.

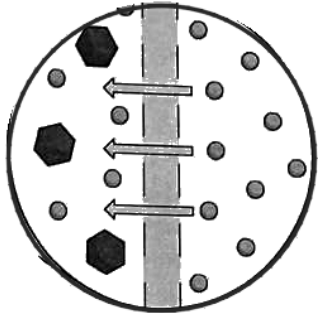
时间 0:33–0:35。

空格: evaporates (is lost) / pulls / xylem / leaves / xylem vessels。

她 booklet p.17 的答案是 “The water in the cylinder are being transport up to the stem. The food colour will change the colour of plant”，只到 A: 没有 xylem，没有原因。

L9: 不能写 “the plant sucks / drinks”，要写 transpiration 造成 pull。

Osmosis



Left: sugar solution. Right: water.

- 1 Which shape is a water molecule?
- 2 Which way do the water molecules move?
- 3 Why can the sugar molecules not cross?

Osmosis is the movement of _____ from a _____ concentration of water to a _____ concentration of water through a _____ membrane.

Teaching line, not a question slide.

时间 0:35-0:37。

1. the small circles (六边形是 sugar molecules)
2. from right to left, into the sugar solution
3. the sugar molecules are too large to fit through the membrane

定义空格: water / high / low / semi-permeable。

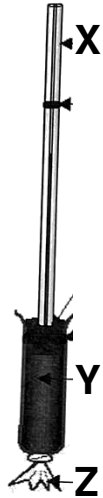
L8: 说 “sugar moves” 或者漏掉 membrane 都会丢分; “high to low” 一定要说是 water 的浓度。

Say it:

semi-permeable = sem-ee-PER-mee-uh-bul

molecule = MOL-ih-kyool

Workbook: the Visking tubing model



The tubing Z sits in a beaker of water. Over 30 minutes the liquid rises up X.

- 1 Which part acts like the root hair? Which acts like the stem?
- 2 Explain why the liquid rises.
- 3 A root hair is long and thin. How does this help it?

1. Z (the Visking tubing) is the root hair; X (the capillary tube) is the stem
2. water moves into the sucrose solution by osmosis, from the beaker (high concentration of water) through the membrane, so the liquid is pushed up the tube
3. it has a large surface area, so it can absorb more water by osmosis

Short solution:

2. Visking tubing 是 semi-permeable membrane: water 可以过, sucrose 不能过

Watch for:

1. 这是 booklet p.18 的 Conclusions, 她没写
3. 只写 “absorbs water” 不够, 要说 surface area

Say it:

sucrose = SOO-krohs

Spot the error (3)

A white carnation stands in blue dye. A day later its petals are blue. Explain.

Sam's answer

The blue water was carried up the phloem tubes. The flower sucks the water up because it is thirsty. Water always gets into a plant by osmosis, from a low concentration of water to a high concentration of water.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. the water goes up the xylem, not the phloem (L7)
2. not “sucks because it is thirsty” : water evaporates through the stomata (transpiration) and pulls water up the xylem (L9)
3. osmosis is from a high to a low concentration of water (L8)

Short solution:

Water evaporates from the flower and leaves through the stomata (transpiration). This pulls water, with the dye, up the xylem, so the petals turn blue. Water enters root hairs by osmosis, from a high to a low concentration of water.

Watch for:

第 3 句只算一个 error (方向反了)。

Practice

Q1 Name the process that moves water into a root hair, and the tubes that carry it to the leaves. **[A]**

Q2 In a Visking tubing model the liquid is 12 mm up the tube at the start, 30 mm after 10 minutes and 48 mm after 20 minutes.

(a) Calculate the average rate of rise in mm per minute.

(b) Explain why the liquid rises. **[M]**

Q3 A plant wilts when its roots are put in very salty water. Explain why. **[E]**

Q1 osmosis; xylem

Q2 (a) 1.8 mm per minute

Q2 (b) water enters the sugar solution by osmosis through the membrane, so the volume rises

Q3 the salty water has a lower concentration of water than the root cells, so water moves out of the roots by osmosis; the plant loses water and wilts

Short solution:

Q2 (a) rise = $48 - 12 = 36$ mm; rate = $\frac{36 \text{ mm}}{20 \text{ min}} = 1.8$ mm per minute

Q3 先比较 water concentration (外面低、里面高)，再说方向 (out of the roots)，最后说结果 (wilts)

Watch for:

Q2 (a) 常见错误：用 48 除以 20 = 2.4，忘了减去开始的 12 mm

Q3 这是 osmosis 换个角度问：水从 high 到 low，所以这次是从根里往外走

Ways to lose marks: Green Machines

Teaching line, not a question slide.

时间 0:38–0:39。

让她每一条说一个今天课上出现过的例子：L1 Spot the error 1；L2 spider plant；L4 和 L6 Spot the error 2；L7 carnation。

L8 和 L9 在作业 hw1 里还会再练。

- L1** Pollination moves pollen; fertilisation fuses the nuclei.
- L2** Asexual: one parent, no gametes, identical offspring.
- L3** Feature **and** how it helps.
- L4** Light and chlorophyll go above and below the arrow.
- L6** Iodine shows starch, not glucose.
- L7** Xylem carries water; phloem carries glucose.

Summary: Green Machines

Teaching line, not a question slide.

时间 0:39–0:40。

让她闭上眼说一遍这四句，再看屏幕对一下。

Flowers: pollination moves pollen; fertilisation joins the nuclei.

Seeds need water, warmth and oxygen; dispersal cuts competition.

Leaves: carbon dioxide + water make glucose + oxygen, using light.

Water: osmosis into root hairs, up the xylem, out by transpiration.

Classwork: Green Machines

On your classwork sheet, about 10 minutes:

A Flowers and seeds

B Photosynthesis and respiration

C Transport and osmosis, with a graph to draw

Write in full sentences for every **explain** question.

Teaching line, not a question slide.

时间 0:40–0:50。

她做的时候老师不要讲，最后 2 分钟对 Part C 的图 (标尺、轴标签、单位、描点)。

答案在 classwork_answers.pdf：Part A 重点是 pollination 和 fertilisation 的区分；Part B 重点是 bell jar 实验的 control；Part C 的图从 0 到 25 分钟是直线上升，之后变平。

时间不够就把 Part C 的画图留到下课后。

LESSON 2 · Metallic materials

Learning intention: link properties to uses, calculate density, and explain alloys, rusting and the reactions of metals.

Teaching line, not a question slide.

时间 0:50–0:51。

这本 booklet 从 p.9 Alloys 往后她几乎没写，学校教过。所以 Loop 2–4 每一页都要让她自己说，不要老师先讲。

四段：Loop 1 properties and density, Loop 2 heat treatment and alloys, Loop 3 corrosion, Loop 4 reactions and the reactivity series。

Say it:

reactivity = ree-ak-TIV-ih-tee

alloys = AL-oyz

corrosion = kuh-ROH-zhun

Do Now

- 1 Give two physical properties shared by most metals.
- 2 Name the three common magnetic metals.
- 3 Write the formula for density.
- 4 Gallium melts at 30 °C. Is it solid or liquid on a 35 °C day?
- 5 A metal is heated red hot, then cooled slowly. Name this heat treatment.

Q1 any two: lustrous, malleable, ductile, sonorous, good conductors of heat and electricity, high melting point

Q2 iron, nickel, cobalt

Q3 density = $\frac{\text{mass}}{\text{volume}}$

Q4 liquid

Q5 annealing

Short solution:

Q4 温度高于 melting point 就是 liquid

Q5 annealing = cool slowly; quenching = cool quickly in cold water

Watch for:

Q1 写 “shiny” 可以，但考试更喜欢 lustrous；写 “strong” 不算通用性质

Q3 写成 mass × volume 是 booklet p.4 她表格里出过的错 (L10)

Diagnostic:

Q3 错：Loop 1 的 density 两页要慢讲

Q5 错：heat treatment 名称不熟，Loop 2 的 Quick review 先做

Say it:

malleable = MAL-ee-uh-bul

ductile = DUK-tile

lustrous = LUS-trus

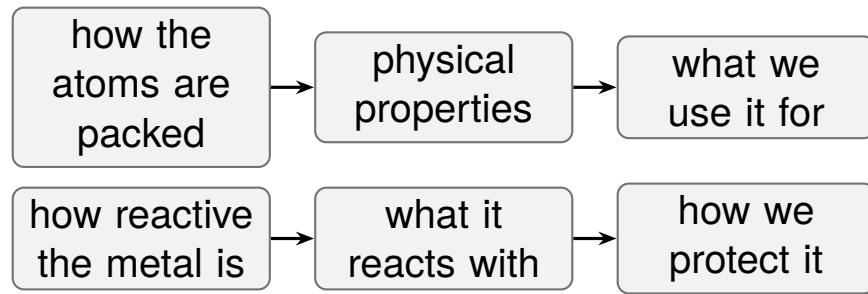
annealing = uh-NEE-ling

gallium = GAL-ee-um

sonorous = SON-uh-rus

quenching = KWEN-ching

The one idea



Top row: physical properties. Bottom row: chemical reactions.
Every Metals question sits on one of these two rows.

Teaching line, not a question slide.

时间 0:54–0:55。

上排: atoms 怎么排列 → physical properties (malleable, conductor, density) → uses。 alloy 也在这一排。

下排: reactivity → 和 oxygen、 water、 acid 的反应 → rusting 和怎么保护。

问她: “为什么 gold 做首饰?” 两排都用到: lustrous (上排) + very unreactive, 不会 corrode (下排)。

Quick review: Property words

Match each word to its meaning, then spell it.

- | | |
|--------------------|---|
| 1 malleable | A has a shiny surface that reflects light |
| 2 ductile | B can be hammered into shape |
| 3 lustrous | C makes a ringing sound when hit |
| 4 sonorous | D can be stretched into wires |
| 5 conductor | E lets heat or electric current pass through |

1. B
2. D
3. A
4. C
5. E

Short solution:

记法: malleable 像 mallet (锤子) ; ductile 像 duct (管子、拉长)

Watch for:

1. 拼写: 两个 l, -eable 结尾 (malleable)
4. sonorous 只有一个 n 在开头

Choosing a metal

Metal	Melting point (°C)	Strength	Electrical conductivity	Density (g/cm ³)
W	1450	high	low	7.8
X	650	low	high	2.7
Y	1080	medium	very high	8.9
Z	230	low	medium	7.3

Which metal for (a) a bridge frame, (b) wiring in a house, (c) a light overhead power cable? (d) Which would melt in a Bunsen flame (about 800 °C)?

Teaching line, not a question slide.

时间 0:57–0:59。

- (a) W: strength high, melting point high
 - (b) Y: electrical conductivity very high
 - (c) X: conductivity high and density low (2.7), 所以 light
 - (d) X 和 Z: melting point 都低于 800 °C
- 这一页先只说选哪个，下一页再练怎么写理由。

Justify a choice: write it in this order

- 1 Name the metal.
- 2 Give the property that matters for this use.
- 3 Say why that property matters for the use.
- 4 Compare: why not another metal? Use the data.

A: 1 + 2 **M:** 1 to 3 **E:** 1 to 4 with numbers

For the power cable I choose _____ because it _____.

This matters because _____.

Y conducts better, but _____.

Teaching line, not a question slide.

时间 0:59–1:01。

E 级范例：I choose X because it has a high electrical conductivity and a low density of 2.7 g/cm^3 . A light cable puts less weight on the pylons and does not sag. Y conducts even better, but its density is 8.9 g/cm^3 , more than three times heavier.

她 booklet p.6 第 5 题写 “Copper, good conductivity and won’t burn you to death”：有 property，但没有 “为什么这个 property 重要”，也没有对比，只有 A (L12)。

Density: the formula and the volume

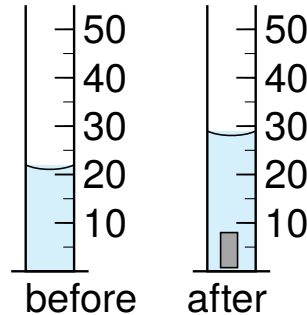
density = $\frac{\text{mass}}{\text{volume}}$

units: g/cm^3 or g/mL

Volume of a regular block:

Volume of an odd-shaped object:
the _____ method

What is the volume of the object on
the right?



Teaching line, not a question slide.

时间 1:01–1:03。

空格: mass / volume; g/mL ; length $\times$ width $\times$ height; displacement.

问题答案: $29 - 22 = 7 \text{ mL}$ (读刻度: before 22 mL, after 29 mL)。

$1 \text{ cm}^3 = 1 \text{ mL}$, 所以 g/cm^3 和 g/mL 是同一个单位。

Say it:

displacement = dis-PLAYS-ment

Density: worked example

A brass key has a mass of 25.5 g. When it is lowered into a measuring cylinder, the water level rises from 30.0 mL to 33.0 mL.

Step 1 volume = _____ – _____ = _____ mL

Step 2 density = _____ ÷ _____

Step 3 density = _____ with its unit

Check: is your answer bigger than 1 g/mL? Brass sinks, so it should be.

Teaching line, not a question slide.

时间 1:03–1:04。

Step 1: $33.0 - 30.0 = 3.0 \text{ mL}$

Step 2: $\frac{25.5 \text{ g}}{3.0 \text{ mL}}$

Step 3: 8.5 g/mL

L11: volume 是两次读数的差，不是最后的读数。L10: 除法，单位 g/mL。

Spot the error (4)

A metal bolt has a mass of 39.5 g. When it is put into a measuring cylinder, the water rises from 20.0 mL to 25.0 mL. Calculate the density of the bolt.

Sam's answer

$$\text{volume} = 25.0 \text{ mL}$$

$$\text{density} = 39.5 \times 25.0 = 987.5 \text{ g}$$

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. volume = $25.0 - 20.0 = 5.0 \text{ mL}$, not 25.0 mL (L11)

2. density = mass divided by volume, not multiplied (L10)

3. the unit is g/mL, not g (L10)

Short solution:

$$\text{volume} = 25.0 - 20.0 = 5.0 \text{ mL}; \text{ density} = \frac{39.5 \text{ g}}{5.0 \text{ mL}} = 7.9 \text{ g/mL} \text{ (和 booklet p.6 表里的 iron 一样)}$$

Watch for:

她 booklet p.4 最后一行也是乘了 (16.55 g 和 2 cm 得出 33.25)，体积单位还写成 cm。答案 987.5 明显不合理，要养成检查“是否合理”的习惯。

Practice

Q1 A metal block is $2.0\text{ cm} \times 2.0\text{ cm} \times 3.0\text{ cm}$ and has a mass of 64.8 g. Calculate its density. **[A]**

Q2 A ring has a mass of 31.5 g. It makes the water in a measuring cylinder rise from 40.0 mL to 43.0 mL. Which metal is it? **[M]**
iron 7.9 copper 8.9 silver 10.5 aluminium 2.7 (g/mL)

Q3 Silver conducts electricity better than copper. Explain why house wiring is made of copper. **[E]**

Q1 5.4 g/cm^3

Q2 silver (10.5 g/mL)

Q3 copper is almost as good a conductor as silver, is much cheaper, and is ductile, so it can be drawn into wires

Short solution:

$$\text{Q1 volume} = 2.0 \times 2.0 \times 3.0 = 12.0\text{ cm}^3; \text{ density} = \frac{64.8\text{ g}}{12.0\text{ cm}^3} = 5.4\text{ g/cm}^3$$

$$\text{Q2 volume} = 43.0 - 40.0 = 3.0\text{ mL}; \text{ density} = \frac{31.5\text{ g}}{3.0\text{ mL}} = 10.5\text{ g/mL}, \text{ 对表得 silver}$$

Q3 E 要有对比: booklet p.6 的数据 silver 60、copper 50, 差别不大, 但 silver 是贵金属

Watch for:

Q2 先减读数再除, 答案要和表对上才算数

Say it:

aluminium = al-yoo-MIN-ee-um

Quick review: Heat treatments

- 1 Heat red hot, then cool quickly in cold water.
- 2 Heat red hot, then cool slowly.
- 3 Heat strongly until a blue sheen appears.

Name each treatment: annealing, quenching or tempering.

Then match it to a result: **A** soft, easy to bend **B** hard but brittle
C tough and springy

- 4 A hair clip must spring back into shape. Which treatment does it need?

1. quenching, B
2. annealing, A
3. tempering, C
4. tempering

Short solution:

4. hair clip 要 springy, 所以是 tempering

Watch for:

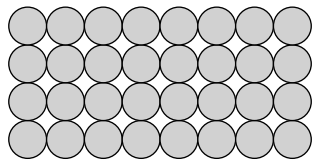
4. 她 booklet p.8 第 2 题写 “Annealing because it modified easily”, 这是错的: annealing 让钢变软, 夹子会弯了回不去

Say it:

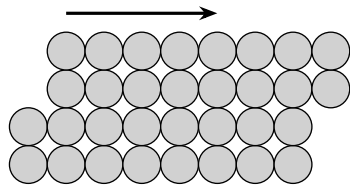
tempering = TEM-per-ing

Why alloys are harder (1)

A pure metal: every atom is the same size, in regular layers.



before



after a push

What did the layers do when they were pushed? So is a pure metal hard or soft?

Teaching line, not a question slide.

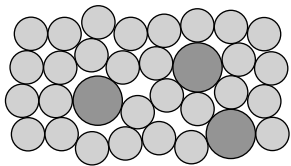
时间 1:07–1:08。

答案: the layers slide over each other easily, 所以 pure metal 是 soft 的, 也因此 malleable 和 ductile。

booklet p.9: pure iron is too soft for many uses.

Why alloys are harder (2)

An alloy: a metal mixed with another element.



- 1 The atoms are _____ sizes.
- 2 This _____ the regular layers.
- 3 The layers cannot _____ easily.
- 4 So the alloy is _____ than the pure metal.

A: harder **M:** + different-sized atoms **E:** whole chain

Teaching line, not a question slide.

时间 1:08–1:09。

空格: different / disrupts / slide over each other / harder。

L13: 只写 “because it is a mixture” 不给分, 必须写 atoms of different sizes 和 layers can't slide。

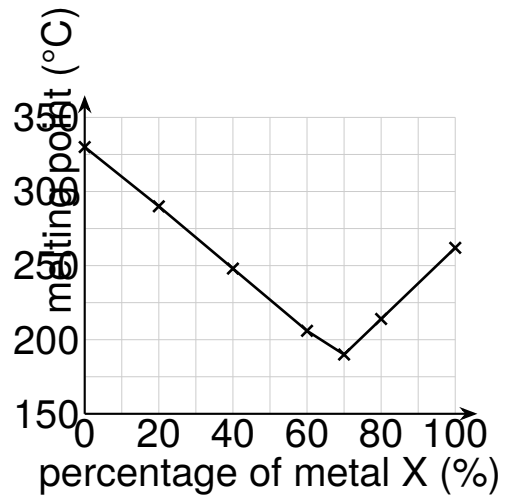
Alloys in use

Alloy	Main metals	Use
bronze	copper + _____	gears, machine bearings
solder	tin + lead	_____
stainless steel	iron, chromium, nickel	_____
nitinol	nickel + titanium	_____

Why is solder good for joining electrical wires?
Is an alloy a compound or a mixture?

Teaching line, not a question slide.
时间 1:09–1:10。
空格：tin / joining electrical wires / cutlery and sinks (resists corrosion) / teeth braces and spectacle frames (shape memory)。
solder：melting point 低，容易熔化，冷却后在常温下是 solid。
alloy 是 mixture：比例可以变，每种成分保留自己的 chemical identity；compound 是固定比例、化学结合。
nitinol 是 smart alloy (shape memory alloy)：弯了以后加热或通电会回到原来的形状。
Say it:
chromium = KROH-mee-um
nitinol = NIT-ih-nol
titanium = ty-TAY-nee-um

Melting points of alloys



Mixtures of two metals, X and Y.

- 1 What is the melting point of pure X? Of pure Y?
- 2 Which mixture has the lowest melting point?
- 3 Describe the trend from 0 % to 70 % X.

Teaching line, not a question slide.

时间 1:10–1:11。

1. pure X (100 %) 262 °C; pure Y (0 % X) 330 °C

2. 70 % X, about 190 °C

3. the melting point falls steadily as the percentage of X increases

提醒 booklet 的 “A SLurP of Tea”: Axes, Scale, Label, Plot, Title。她 p.10 的 solder 图还没画，留在作业里。

结论 (和 booklet p.11 Q1 同一个思路)：alloy 的 melting point 可以比两种纯金属都低。

Practice

Q1 Name the heat treatment that makes steel hard but brittle. **[A]**

Q2 Brass is an alloy of copper and zinc. Explain why brass is harder than pure copper. **[M]**

Q3 A car spring and a car door panel are both made of steel. Explain which heat treatment each one should have. **[E]**

Q1 quenching

Q2 zinc atoms are a different size from copper atoms; they disrupt the regular layers, so the layers cannot slide over each other easily

Q3 spring: tempering, so it is tough and springy and returns to shape; door panel: annealing, so it is soft and easy to press into shape

Short solution:

Q3 每一个都要 “treatment + 得到的 property + 为什么这个用途需要它”

Watch for:

Q2 只写 “alloys are harder” 是复述题目，0 分

Q3 门板写 quenching 是错的：太 brittle，一撞就裂

Quick review: Corrosion and rusting

True or false?

- 1 Rusting is a type of corrosion.
- 2 Any metal can rust.
- 3 Salt water makes iron rust faster.
- 4 Gold corrodes quickly in air.
- 5 Corrosion is a chemical reaction of a metal with substances such as oxygen and water.

1. true
2. false
3. true
4. false
5. true

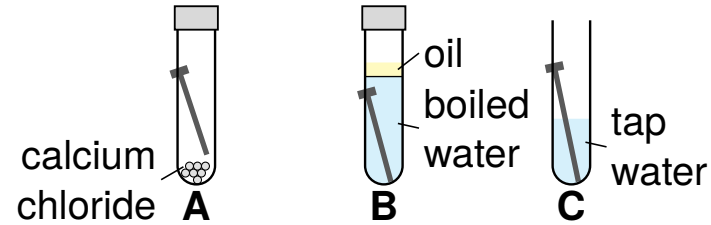
Short solution:

2. 只有 iron (和 steel) 会 rust; 其他金属是 corrode
3. booklet p.13: salt solution 和 higher temperature 都会加快 rusting

Watch for:

LO 要求 “distinguish between corrosion and rusting”: rusting 是 corrosion 的一种, 只用于 iron

Rusty nail experiment (1): the set-up



Iron nails, left for 3 days.

A The calcium chloride removes _____.

B Boiling removes _____ from the water. The oil stops _____.

C The nail has both _____ and _____.

Teaching line, not a question slide.

时间 1:12–1:14。

A: water (moisture) from the air, 所以 A 是 dry air

B: dissolved oxygen (air); air from dissolving back into the water

C: water and oxygen (air)

这个设计和 Lesson 1 Practice Q3 的种子实验一样: boiled water + oil = 没有 oxygen。

Say it:

calcium = KAL-see-um

Rusty nail experiment (2): write it in this order

Results after 3 days: A _____ B _____ C _____

- 1 Only tube ____ rusted.
- 2 Tube A had air but no _____.
- 3 Tube B had water but no _____.
- 4 So iron needs _____ to rust.

A: C rusted **M:** + what A and B lacked **E:** the conclusion from all three

Teaching line, not a question slide.

时间 1:14–1:15。

结果：A no rust; B no rust; C rust。

空格：C / water / oxygen / both water and oxygen。

她 booklet p.17 的结果写 B “slight rusting, barely can see”，是错的。B 没有 oxygen，不应该生锈；如果真的有一点，说明 boiled water 又溶进了空气 (L14)。

Spot the error (5)

Steel paper clips are sealed in three tubes: P with calcium chloride (dry air), Q in boiled water under a layer of oil, R in tap water with air above. Describe and explain the results.

Sam's answer

P: no rust, because there is no air. Q: a little rust, because it has water. R: lots of rust. So iron only needs water to rust.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. P has air; it does not rust because it has no water (L14)
2. Q does not rust: boiled water has no oxygen and the oil keeps air out (L14)
3. iron needs both water and oxygen to rust (L14)

Short solution:

P: no rust, because the calcium chloride removes the water. Q: no rust, because boiling removes the oxygen and the oil keeps air out. R: lots of rust, because it has water and oxygen. So iron needs both water and oxygen to rust.

Watch for:

R 那一句是对的。steel 也会 rust, 因为 steel 主要是 iron。

Preventing rust

Object	Method	Why this method
steel fence	_____	large, cheap, easy to redo
door hinge	oil or grease	_____
rubbish bin	_____	zinc coating
kitchen sink	_____	never needs a coating

Every method keeps _____ and _____ away from the iron.

Teaching line, not a question slide.
时间 1:16–1:17。
空格：painting；moving parts，paint would rub off；galvanising；made of stainless steel (alloying)。
最后一句：water and oxygen (air)。
booklet p.13：选哪种方法看 object 的大小、用途和成本。
Say it:
galvanising = GAL-vuh-ny-zing

Practice

Q1 Name the two substances iron needs in order to rust. **[A]**

Q2 Explain why a car in a seaside town rusts faster than the same car far inland. **[M]**

Q3 Plan an experiment to find out whether salt water makes iron nails rust faster than tap water. Say what you would change, what you would measure, and what you would keep the same. **[E]**

Q1 water and oxygen

Q2 the air near the sea carries salt, and salt solution speeds up rusting

Q3 change: tap water or salt water; measure: how much rust after 3 days; keep the same: type and size of nail (cleaned with sandpaper), volume of water, temperature, time, tubes open to the air; repeat each tube

Short solution:

Q3 E 要有: independent variable、dependent variable、至少三个 controlled variables、重复实验

Watch for:

Q3 她实验题最容易漏 “how to measure”: 写 “compare the amount of rust” 或 “measure the mass of the nail before and after” 都可以

Quick review: Word equation patterns

- 1 metal + oxygen → _____
- 2 metal + water → _____ + _____
- 3 metal + hydrochloric acid → _____ + _____
- 4 Test for hydrogen: a lit splint gives a _____.

1. metal oxide
2. metal hydroxide + hydrogen
3. metal chloride + hydrogen
4. squeaky pop

Short solution:

2 和 3 都放出 hydrogen gas, 所以都可以用 lit splint 测

Watch for:

2. 写成 metal oxide + hydrogen 是常见错误 (L15)

Say it:

hydroxide = hy-DROK-side

hydrochloric = hy-droh-KLOR-ik

Writing word equations

(a) Iron wool burns in oxygen.

iron + oxygen → _____

(b) Lithium fizzes in cold water.

lithium + water → _____ + _____

(c) Aluminium powder in dilute hydrochloric acid.

(d) Gold in dilute hydrochloric acid.

Teaching line, not a question slide.

时间 1:21–1:23。

(a) iron oxide

(b) lithium hydroxide + hydrogen

(c) aluminium + hydrochloric acid → aluminium chloride + hydrogen

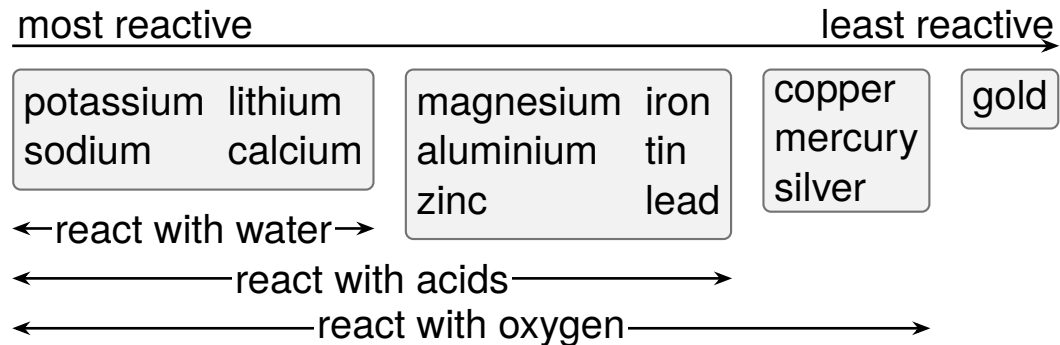
(d) no reaction: gold 在 reactivity series 的最下面

考试写 word equation: 用词, 不用化学式; 箭头不要写成等号。

Say it:

lithium = LITH-ee-um

The reactivity series



1 Would tin react with dilute acid?

2 Why is gold found in the ground as pure metal, but iron is not?

Teaching line, not a question slide.

时间 1:23–1:25。

1. yes: tin 在 “react with acids” 的范围里 (反应很慢)

2. gold is so unreactive that it does not react with oxygen or sulfur, so it stays as the pure metal; iron is more reactive, so it is found as compounds (ores) and must be extracted

这一页对应 booklet p.20 Summary 的图，重画成四组。

Say it:

potassium = puh-TAS-ee-um

magnesium = mag-NEE-zee-um

mercury = MER-kyuh-ree

Spot the error (6)

Complete each word equation, or write “no reaction”.

Sam’s answer

1. calcium + oxygen makes calcium oxygen
2. iron + hydrochloric acid makes iron chloride + water
3. potassium + water makes potassium hydroxide + hydrogen
4. silver + hydrochloric acid makes silver chloride + hydrogen

Sam’s answer has 3 errors. Find each one, fix it, and say why it loses marks.

1. calcium + oxygen → calcium oxide, not calcium oxygen (L15)
2. iron + hydrochloric acid → iron chloride + hydrogen, not water (L15)
3. silver + hydrochloric acid → no reaction: silver is below the acids in the series (L16)

Short solution:

calcium + oxygen → calcium oxide; iron + hydrochloric acid → iron chloride + hydrogen; potassium + water → potassium hydroxide + hydrogen (correct); silver + hydrochloric acid → no reaction

Watch for:

第 3 条是对的，error 的编号按找到的顺序 1-3，不按 Sam 的行号。

Practice

Q1 Write the word equation for aluminium reacting with oxygen. **[A]**

Q2 Four metals are added to dilute acid. Put them in order of reactivity, most reactive first, and give your evidence. **[M]**

P many bubbles, quickly

Q no bubbles

R a few bubbles, slowly

S reacts even with cold water

Q3 Describe an experiment with dilute hydrochloric acid to show that zinc is more reactive than copper. Give the result you expect. **[E]**

Q1 aluminium + oxygen → aluminium oxide

Q2 S, P, R, Q

Q3 add same-sized pieces of zinc and copper to the same volume of the same dilute hydrochloric acid; zinc fizzes (hydrogen), copper does not react

Short solution:

Q2 evidence: S 连冷水都能反应, 最活泼; P 气泡多又快; R 少又慢; Q 没有反应

Q3 fair test: same size, same volume, same concentration, same temperature; 可以用 lit splint 证明气体是 hydrogen

Watch for:

Q2 只排顺序不写证据只有 A (L16)

Ways to lose marks: Metals

L10 Density = mass $\div$ volume, unit g/mL or g/cm³.

L11 Volume = final reading – first reading.

L12 Property + why it matters + comparison.

L13 Different-sized atoms, layers cannot slide.

L14 Rusting needs water **and** oxygen.

L15 Oxide; hydroxide + hydrogen; chloride + hydrogen.

Teaching line, not a question slide.

时间 1:27–1:28。

L10 和 L11 来自她 booklet p.4 的表； L14 来自 p.17 的 tube B； L12 来自 p.6。

L16 在作业 hw2 里再练。

Summary: Metallic materials

Teaching line, not a question slide.

时间 1:28–1:29。

同样让她先说再看。

Properties decide uses: say the property, why it matters, and compare.

Density = mass $\div$ volume; displacement gives the volume.

Alloys are harder: different-sized atoms stop the layers sliding.

Rust needs water and oxygen; reactivity decides how metals react.

Classwork: Metallic materials

On your classwork sheet, about 10 minutes:

D Properties and density

E Alloys and heat treatment

F Corrosion and reactions, with an experiment to plan

Show every step of a calculation, with units.

Teaching line, not a question slide.

时间 1:29–1:40。

答案在 classwork_answers.pdf: Part D 的 density 题答案是 2.7 g/cm^3 (aluminium); Part E 重点是 alloy 的解释链; Part F 的实验设计要有三个 controlled variables。

1:38 收卷, 准备发 Mini mock。

Mini mock

17 minutes · both units · on the paper

Answer every part.

Each question is marked A, M or E.

Explain questions: write it in the order you practised today.

Teaching line, not a question slide.

时间 1:40–1:57。

发 mock.pdf，计时 17 分钟，老师不提示。

答案和评分标准在 mock_answers.pdf，每题写了 A、M、E 各需要写到哪一步。

时间到了先收卷，课后批改，下次课开头花 5 分钟讲最差的两题。

Homework

hw1 Green Machines, about 40 minutes

hw2 Metallic materials, about 40 minutes

Also: Metals booklet p.10–11, draw the solder graph and answer Q1–2.

Do them on two different days before the exam.

Teaching line, not a question slide.

时间 1:57–2:00。

hw1: pollination、osmosis、transpiration 的 Explain 题和一个 Spot the error; hw2: density 计算、reactivity 数据题、rusting 实验设计和一个 Spot the error。

答案文件 hw1_answers.pdf 和 hw2_answers.pdf，同版面红色答案。

booklet p.10 的 solder 图：提醒 SLurP of Tea，x 轴是 percentage of tin。